

AISI 1005 Steel

Categories: [Metal](#); [Ferrous Metal](#); [Carbon Steel](#); [AISI 1000 Series Steel](#); [Low Carbon Steel](#)

Material Notes: AISI 1005 Low Carbon Steel

Key Words: UNS G10050, AISI1005, ASTM A29, ASTM A510, MIL SPEC MIL-S-11310 (CS1005), SAE J403, SAE J412

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	7.872 g/cc	0.2844 lb/in ³	Composition 0.06% C, 0.38% Mn, 0.01% Si, annealed at 925°C

Mechanical Properties	Metric	English	Comments
Modulus of Elasticity	200 GPa	29000 ksi	Typical for steel
Bulk Modulus	160 GPa	23200 ksi	Typical for steel
Poissons Ratio	0.29	0.29	Typical For Steel
Shear Modulus	80.0 GPa	11600 ksi	Typical for steel

Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.0000174 ohm-cm	0.0000174 ohm-cm	Typical steel

Thermal Properties	Metric	English	Comments
CTE, linear 	12.6 µm/m-°C	7.00 µin/in-°F	
	@Temperature 0.000 - 100 °C	@Temperature 32.0 - 212 °F	
	13.1 µm/m-°C	7.28 µin/in-°F	
	@Temperature 0.000 - 200 °C	@Temperature 32.0 - 392 °F	
	13.5 µm/m-°C	7.50 µin/in-°F	
	@Temperature 0.000 - 300 °C	@Temperature 32.0 - 572 °F	
	13.7 µm/m-°C	7.61 µin/in-°F	
	@Temperature 0.000 - 400 °C	@Temperature 32.0 - 752 °F	
	13.7 µm/m-°C	7.61 µin/in-°F	
	@Temperature 0.000 - 1000 °C	@Temperature 32.0 - 1830 °F	
Specific Heat Capacity 	14.2 µm/m-°C	7.89 µin/in-°F	
	@Temperature 0.000 - 500 °C	@Temperature 32.0 - 932 °F	
	14.6 µm/m-°C	8.11 µin/in-°F	
	@Temperature 0.000 - 600 °C	@Temperature 32.0 - 1110 °F	
	14.9 µm/m-°C	8.28 µin/in-°F	
	@Temperature 0.000 - 700 °C	@Temperature 32.0 - 1290 °F	
	16.6 µm/m-°C	9.22 µin/in-°F	
	@Temperature 0.000 - 800 °C	@Temperature 32.0 - 1470 °F	
	0.481 J/g-°C	0.115 BTU/lb-°F	
	@Temperature >=100 °C	@Temperature >=212 °F	
	0.519 J/g-°C	0.124 BTU/lb-°F	
	@Temperature 150 - 200 °C	@Temperature 302 - 392 °F	
	0.536 J/g-°C	0.128 BTU/lb-°F	
	@Temperature 200 - 250 °C	@Temperature 392 - 482 °F	
	0.553 J/g-°C	0.132 BTU/lb-°F	
	@Temperature 250 - 300 °C	@Temperature 482 - 572 °F	
	0.595 J/g-°C	0.142 BTU/lb-°F	
	@Temperature 350 - 400 °C	@Temperature 662 - 752 °F	

0.662 J/g-°C @Temperature 450 - 500 °C	0.158 BTU/lb-°F @Temperature 842 - 932 °F
0.754 J/g-°C @Temperature 550 - 600 °C	0.180 BTU/lb-°F @Temperature 1020 - 1110 °F
0.846 J/g-°C @Temperature 850 - 900 °C	0.202 BTU/lb-°F @Temperature 1560 - 1650 °F
0.867 J/g-°C @Temperature 650 - 700 °C	0.207 BTU/lb-°F @Temperature 1200 - 1290 °F
1.105 J/g-°C @Temperature 700 - 750 °C	0.2641 BTU/lb-°F @Temperature 1290 - 1380 °F

Component Elements Properties	Metric	English	Comments
Carbon, C	<= 0.060 %	<= 0.060 %	
Iron, Fe	99.5 - 100 %	99.5 - 100 %	As remainder
Manganese, Mn	<= 0.35 %	<= 0.35 %	
Phosphorus, P	<= 0.040 %	<= 0.040 %	
Sulfur, S	<= 0.050 %	<= 0.050 %	

[References](#) for this datasheet.

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.